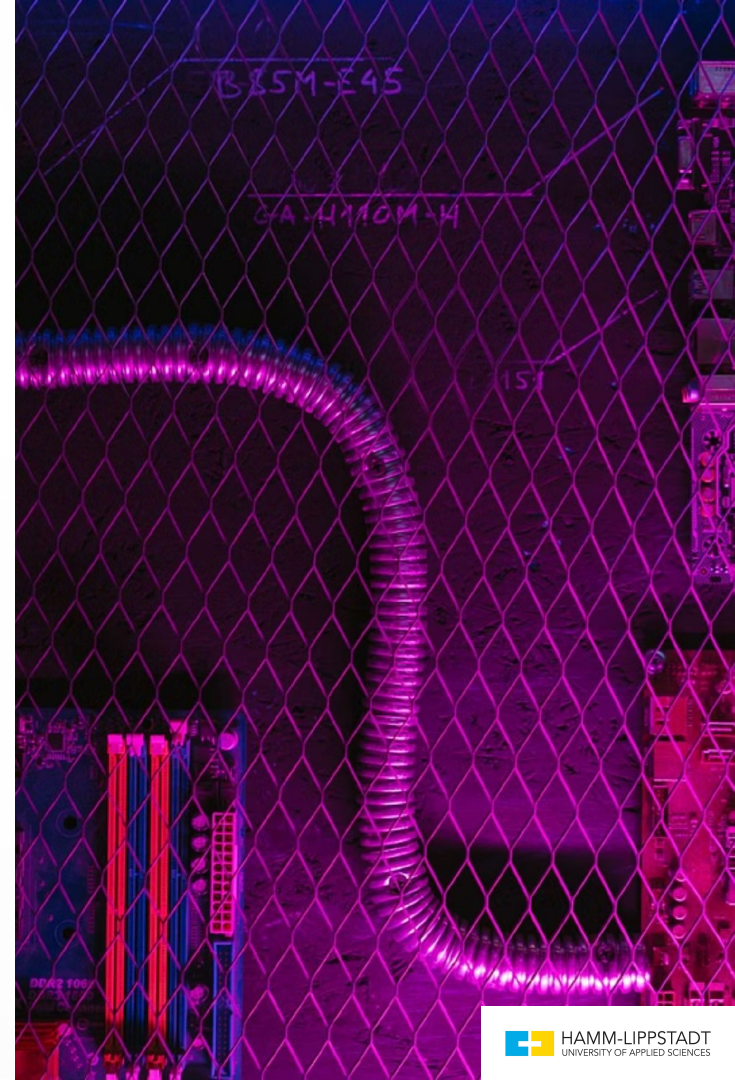


High-Level Synthesis with Ant Colony Optimization Scheduling

A Critical Analysis of Kopuri & Mansouri, IEEE ISCAS 2004

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The Scheduling Problem

HLS transforms behavioural HDL into an RTL netlist. Scheduling = assigning operations to clock cycles. **NP-complete**; shapes every downstream cost decision.

Time Frame

Each operation constrained to [ASAP, ALAP] — the window in which it must be scheduled.

Concurrency → Area

Concurrent ops at the same step need separate hardware units → more area.

FDS Baseline

Paulin & Knight 1987: near-optimal but $O(N^3)$, single-pass, no synthesis feedback.



FDS Baseline: The Force Equations

FDS schedules the (operation, time-step) pair with the **lowest total force**.

Eq. 1 — Self-Force

$$S_{il} = \sum_{m=t_i^S}^{t_i^L} q_k(m)(\delta_{lm} - p_i(m))$$

Eq. 2 — PS-Force

$$PS_{il} = \frac{1}{\tilde{\mu}_i + 1} \sum_{m=\tilde{t}_i^S}^{\tilde{t}_i^L} q_k(m) - \frac{1}{\mu_i + 1} \sum_{m=t_i^S}^{t_i^L} q_k(m)$$

$q_k(l)$

Expected type-k
concurrency at step l

$p_i(m)$

= 1 / (ALAP - ASAP + 1),
uniform probability
over time frame

$O(N^3)$ Bottleneck

PS-force visits all predecessors/successors

ACO Modification: Replace PS-Force

Replace the expensive PS-force with two matrices **learned from real synthesis results**.

Eq. 5 — Modified Force

$$F_{ij} = \alpha \frac{S_{ij}}{\sqrt{\frac{1}{N\lambda} \sum S_{ij}}} - \beta M_{ij}^n - \gamma G_{ij}^n$$

All terms **$O(N\lambda)$** — down from $O(N^3)$ for PS-force. Cost evaluated *after* full synthesis (clique partition + Left-Edge register allocation).

Term 1

Normalised self-force ($\alpha=1$)

Term 2

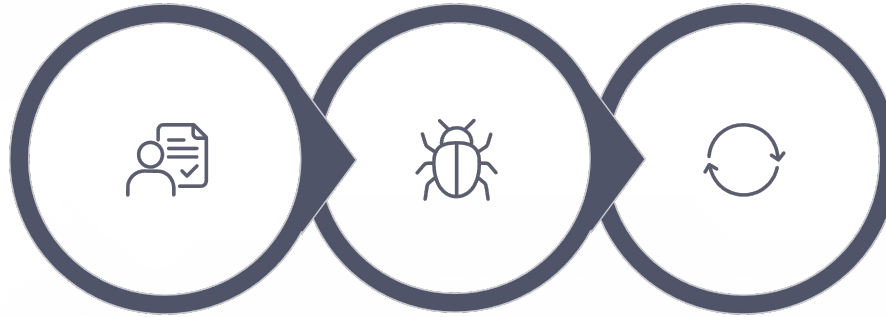
Per-cycle trail ($\beta=2$)

Term 3

Global experience ($\gamma=2$)

Algorithm Ant_sched

20 iterations × 20 ants = **400 synthesis evaluations**. $\eta=0.5$, $\rho=0.35-0.50$



Assign Ants

Run A_FDS

Update G

Eq. 3 — Per-Cycle Trail

$$M_{ij}^n = M_{ij}^{n-1} + \frac{\text{cost}(s) - \text{cost}(s^0)}{\text{cost}(s^0)}, \quad \forall (i,j) \in s$$

Eq. 4 — Global Experience

$$G_{ij}^{n+1} = \rho G_{ij}^n + M_{ij}^n$$

Sign-Convention Inconsistency

⚠ **Eq. 3 as printed:** if s IMPROVES over s^o , $\text{cost}(s) < \text{cost}(s^o) \rightarrow$ increment is **NEGATIVE**. In Eq. 5: $-\beta \cdot M = -2 \cdot (-0.10) = +0.20 \rightarrow$ force INCREASES $\rightarrow A_FDS$ picks MINIMUM $\rightarrow$ improved decision **LESS preferred**.

This is **anti-reinforcement** — the opposite of ACO pheromone logic. Paper prose (p. V-259): "trail is positive or negative depending on whether improved results are obtained." The math always gives negative for improvement. **Contradiction.**

Fix A

Negate Eq. 3

Fix B

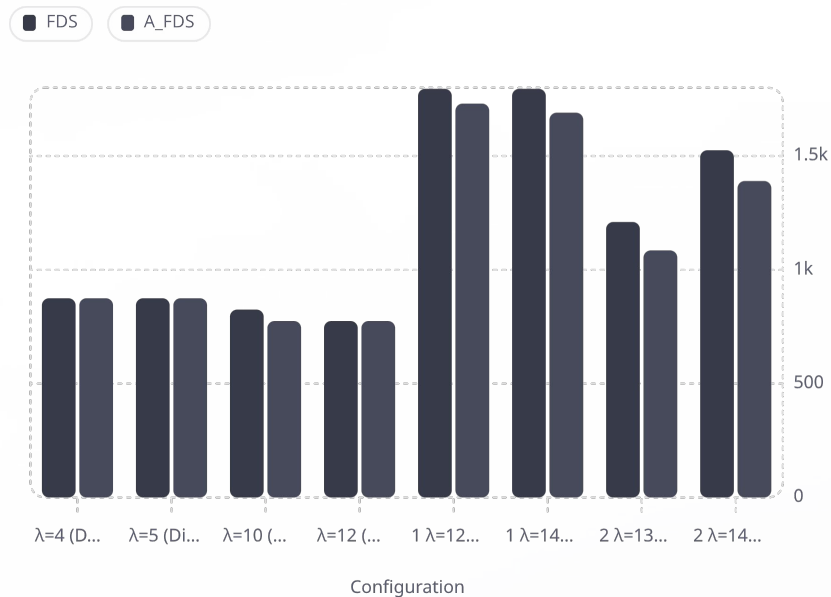
Use $+\beta \cdot M + \gamma \cdot G$ in Eq. 5 and select *maximum*

Why Benchmarks Still Improve

Self-force dominates; trail diversifies search

Benchmark Results

Hardware cost — lower is better. Cost model: Multiplier=250, Adder=50, Register=15, Mux=15. $A_FDS \leq FDS$ in all 8 cases — **never worse**.
Single run per configuration: no statistical variance.



Best Gain

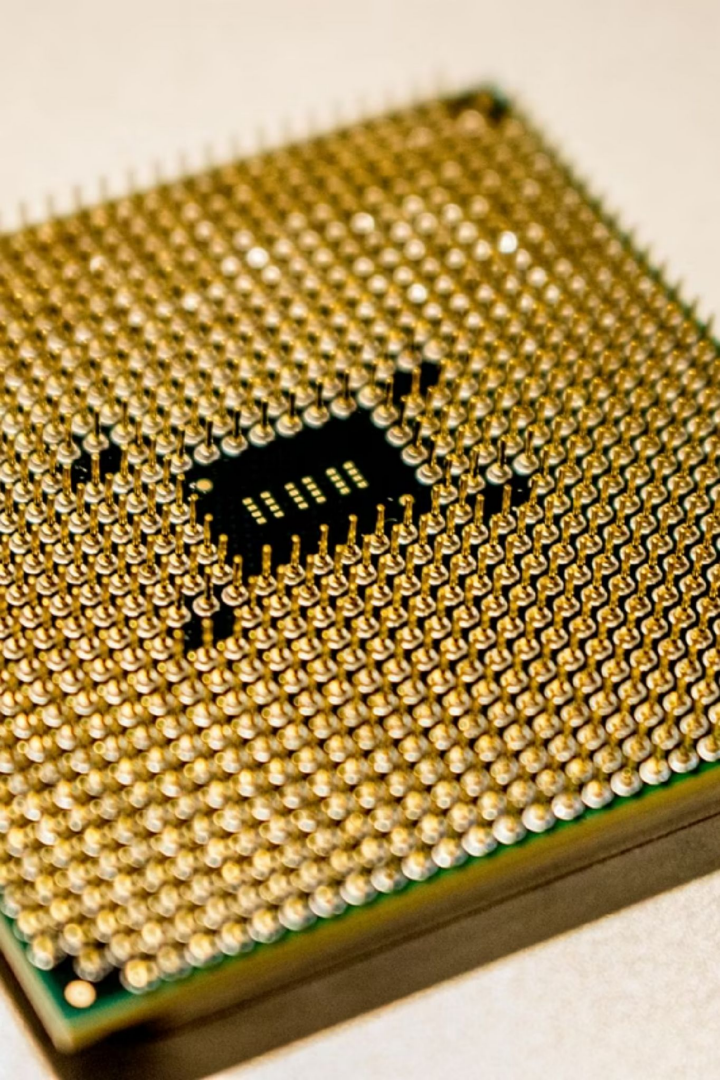
DCT 2 at $\lambda=13$: **-10.3%** (1210 $\rightarrow$ 1085)

No Gain

Differential Equation ($\lambda=4, 5$) and Elliptical Filter ($\lambda=12$): 0% improvement

Runtime

"A few seconds" on Pentium 3 / 800 MHz, 320 MB RAM



Suitability for Embedded Systems



ASIC DSP Datapaths

Fixed-latency designs; area-optimisation plug-in for filter cores, transform engines, linear algebra accelerators.



HLS Toolchain Integration

Requires no modification to resource allocation or register allocation steps.



FPGA Targets $\times$

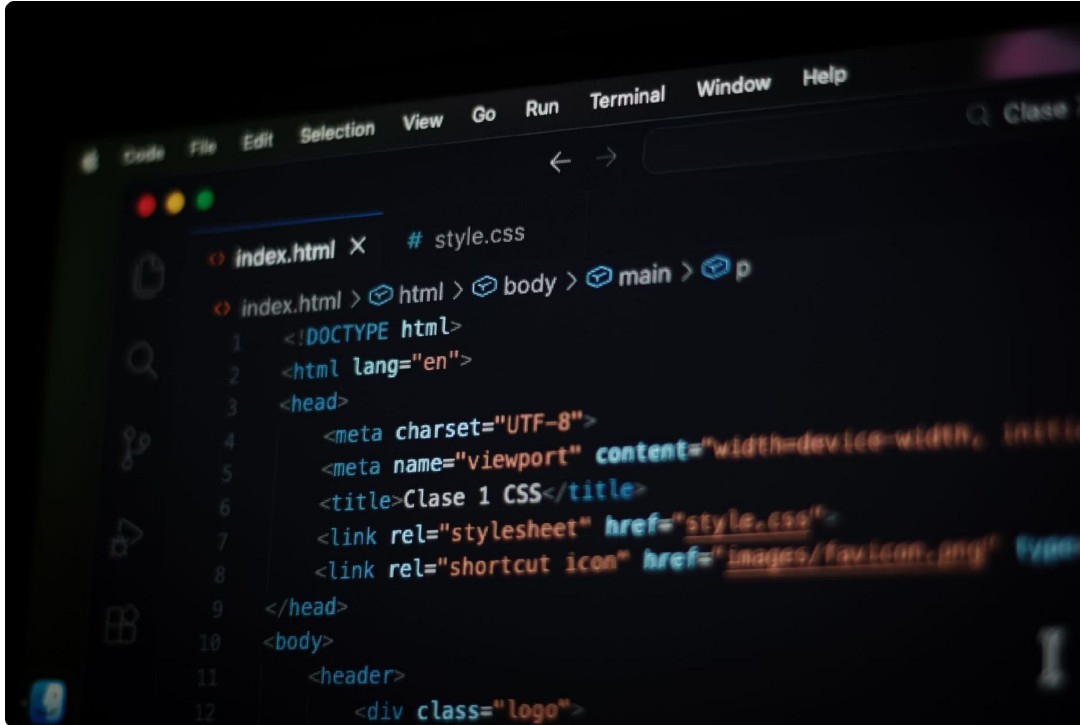
Cost model is count-based; must account for LUT packing and routing congestion instead.



Hard Real-Time $\times$

Stochastic scheduler cannot provide WCET bounds. Control-flow-heavy and memory-bandwidth-bound designs unsuitable.

C++17 Implementation & VHDL Output



Implementation Highlights

- CDFG as adjacency list with ASAP/ALAP annotations
- Trail matrices M^n and G :
`std::vector<double>`, size $N \times \lambda$
- RMS normalisation computed once per A_FDS invocation
- Clique-partitioning resource allocation
+ Left-Edge register allocation
- `fix_sign` flag: corrected sign
converges faster on DCT benchmarks

VHDL RTL Output

Generated entity: 2 multipliers, 2 adders, 6 registers, FSM controller. Synthesisable RTL corresponding to HLS output level.

Conclusion

1 Complexity Reduced

A_FDS cuts per-step cost from $O(N^3)$ to $O(N\lambda)$, feeding real synthesis cost back into scheduling.

2 0–10.3% Cost Reduction

$A_FDS \leq FDS$ across all 8 DSP benchmark configurations within 20 iterations.

3 Sign-Convention Bug

Eq. 3 anti-reinforces improved decisions as printed. Two fixes identified; corrected version converges faster in C++ experiments.

Future work: Statistical validation of both sign conventions · multi-objective cost (area + timing + power) · parallel ants · FPGA cost model

S. Kopuri & N. Mansouri, *IEEE ISCAS 2004*, pp. V-257–V-260